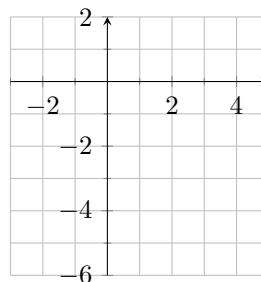


1. Let $\mathbf{x} = (2, -5)$ and $\mathbf{y} = (0, 3)$. Complete the following:

a. $\mathbf{x} + \mathbf{y}$



b. Sketch $\mathbf{x} + \mathbf{y}$

c. Sketch $\mathbf{x} - \mathbf{y}$.

d. Sketch $\frac{1}{2}\mathbf{x}$.

2. Determine if the point $(-6, 2)$ is on the line defined parametrically by $\mathbf{x} = (0, 1) + t(-3, 1)$.

3. Let $(1, 1, 1)$ and $(2, 0, 1)$ be two points on a plane H . Find a line on the plane.

4. Let $\mathbf{x}$ and $\mathbf{y}$ be two vectors such that $\mathbf{x} \cdot \mathbf{y} = 0$.

a. What is the angle between $\mathbf{x}$ and $\mathbf{y}$?

b. If $\mathbf{x} = (1, 1)$ find a vector parallel to $\mathbf{y}$.